



Tara Chand
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B.Tech. - Civil Engineering
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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech.	Indian Institute of Technology, Guwahati	8.55 (Current)	2023-Present
Senior Secondary	Central Board of Secondary Education	94.4%	2022
Secondary	Board of School Education, Haryana	92.8%	2020

EXPERIENCE

- Outsyde - Adventure Booking Platform** May 2026 – Jul 2026
Full-Stack Developer Intern Remote
 - Designed a **51-table PostgreSQL** database with **Prisma**, connecting treks, trips, bookings, payments, referrals, and payouts. Used relational constraints, price and address snapshots, soft deletes, and **append-only audit logs** to maintain data accuracy.
 - Built a **9-section admin console** with role-based permissions across **64+ APIs**, covering provider KYC, user management, refunds, disputes, commissions, and payouts. Fixed a payment race condition using **row-level locking** to prevent duplicate refunds.
 - Developed the booking platform with consumer and provider portals, **retry-safe payments** with duplicate-charge protection, and a loyalty wallet system. Added discovery features such as fuzzy search, trending rankings, and sponsored listings.

PROJECTS

- Smart Traffic Command Center** | *Flipkart Gridlock 2.0 Hackathon* May 2026 | Website | GitHub
 - Engineered **64 key features** from **8,173** traffic incidents and trained gradient-boosting models, achieving **0.989 ROC-AUC**.
 - Formulated a data-driven severity score using **PCA-weighted blending** across 7 indicators, generating actionable recommendations.
 - Developed a **Streamlit** dashboard with a map predictor, model thresholds, and prediction history using reliable crash-safe writes.
 - Established a leakage-safe pipeline using clusters, priority encodings, and training-fitted scalers, ensuring reliable model evaluation.
- Verity - Business Intelligence System** | *Accenture Innovation Challenge* Mar 2026 - Apr 2026 | GitHub
 - Developed a business intelligence system to detect important metric changes using **STL and Isolation Forest** for anomaly detection.
 - Implemented **Retrieval-Augmented Generation** to retrieve key business documents based on the metric, region, and user access.
 - Integrated a **Large Language Model** to explain changes from verified data, with a citation guard preventing unsupported numbers.
 - Designed a decision system with **forecasting** and data analysis across 4 scenarios for key business decisions, with approval checks.
- Hangman-Solving AI: BiLSTM** | *Group Project* Aug 2026 | Website | GitHub
 - Developed an automated Hangman agent using two **Bidirectional LSTM** models trained on **227K+ words** to predict next letters.
 - Created a leak-free **97.5%** and **2.5% train-test split** with fixed seeding and overlap checks for performance on unseen words.
 - Combined model predictions with **information gain and letter frequency** to select effective guesses from partially revealed words.
 - Achieved a **56.2% win rate** on 5,000 held-out words under a strict 6-try limit, averaging 4.28 incorrect guesses per game overall.
- Weed Detection** | *Group Project* April - June 2025 | GitHub
 - Developed **YOLO11m**-based weed detection using labeled and unlabeled agricultural imagery for robust weed localization.
 - Implemented **semi-supervised learning** with FixMatch, pseudo-labeling, and consistency regularization to leverage unlabeled data.
 - Utilized image and bounding-box augmentations to enhance model generalization across diverse agricultural field environments.
 - Achieved **91.75% precision**, **85.05% recall**, and 85% mean average precision overall, demonstrating strong detection performance.

TECHNICAL SKILLS

- Data Analysis and Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau*
 - ML Frameworks:** Scikit-learn, Keras, TensorFlow, LangChain*
 - Databases:** MySQL, MongoDB, PostgreSQL, ChromaDB
 - Programming Languages:** Python, SQL, C++, C
 - Tools and Technologies:** Jupyter, Streamlit, Git, GitHub
- * Elementary proficiency

KEY COURSES TAKEN

- Mathematics:** Linear Algebra and Multivariable Calculus, Complex Analysis, Real Analysis
- Computer Science:** Introduction to Computing, Computing Laboratory

ACHIEVEMENTS

- Silver Flag Winner**, Selected in the **Top 75** among **30K+** registrations for the Cisco Code-a-thon finals, Bangalore 2026
- Top 1.9%**, MyCareernet HackVega 2.0, national coding challenge among **47K+** participants 2026
- Top 4.4%**, Flipkart Gridlock 2.0, ML and Data Science hackathon, among **35K+** participating teams 2026
- Semifinalist**, Flipkart GRiD 8.0, Software Development Track, among **150K+** participants nationwide 2026
- Semifinalist**, Economic Times AI Hackathon 2.0, national AI innovation challenge, among **60K+** participants 2026
- Competitive Programming**, Solved **1200+** DSA problems across Codeforces, CodeChef, LeetCode & AtCoder Link

EXTRA-CURRICULAR ACTIVITIES

- Student Mentor at Physics Wallah**, Guided **750+** JEE aspirants through study plans and weekly doubt-solving 2024
- Participant**, Participated in **Meta Hacker Cup**, a global competitive programming contest 2024
- NSS Volunteer**, Contributed **80+** hours to community service and awareness campaigns at IIT Guwahati 2024
- Participant**, Participated in Mime Battle at Manthan, the Inter-Hostel Cultural Competition at IIT Guwahati 2024
- Core Team**, Contributed to Public Relations, branding, and the management of participants for Alcheringa 2026